## Measurable sets and functions

**Definition 1 ($\sigma$-algebras)** *Let $\Omega$ be a set and $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ be a collection of subsets of $\Omega$:*
*i. $\mathcal{A}$ is an* algebra *if $\varnothing \in \mathcal{A}$ and for $A, B \in \mathcal{A}$, $A^c = \Omega \setminus A \in \mathcal{A}$ and $A \cup B \in \mathcal{A}$.*
*ii. $\mathcal{A}$ is a $\sigma$-*algebra *if $\varnothing \in \mathcal{A}$, for $A \in \mathcal{A}$, $A^c \in \mathcal{A}$, and for $(A_n)$ a sequence of sets in $\mathcal{A}$, $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$.*

A collection of sets is an algebra subject to being closed under finite applications of the basic operators. The $\sigma$-algebra concept extends this slightly to infinite ones. Consider where this distinction is relevant?

Note that if we have $\{\mathcal{F}_i : i \in I\}$ are $\sigma$-algebras, then

$$\mathcal{F} = \bigcap_{i \in I} \mathcal{F}_i$$

is a $\sigma$-algebra. This allows us to consider the notion of a smallest $\sigma$-algebra containing a set (the $\sigma$-algebra 'generated' by a set). We write the $\sigma$-algebra generated by a collection of collections of sets $\mathfrak{A}$ as $\sigma(\mathfrak{A})$.

**Definition 2 (Borel $\sigma$-algebra)** *Let $(E, \mathcal{T})$ be a topological space. The $\sigma$-algebra generated by the open sets in $E$ is called the* Borel $\sigma$-algebra *on $E$ and is denoted $\mathcal{B}(E) = \sigma(\mathcal{T})$.*

**Definition 3 (Product space)** *Suppose $(\Omega_i, \mathcal{F}_i)_{i \in I}$ are measurable spaces. With $\Omega = \prod_{i \in I} \Omega_i$, $\mathcal{F}$ the $\sigma$-algebra generated by $A = \prod_{i \in I} A_i$ where $A_i \in \mathcal{F}_i$ for all $i \in I$ and for all but finitely many $i \in I$, $A_i = \Omega_i$: $(\Omega, \mathcal{F})$ is the product space.*

This space is measurable, and $\mathcal{F}$ is a $\sigma$-algebra.

**Definition 4 ($\pi$ and $\lambda$-systems)** *A collection of sets $\mathcal{A}$ is called a $\pi$-system if it is closed under intersections.*

*A collection of sets $\mathcal{M}$ is called a $\lambda$-system if $\Omega \in \mathcal{M}$, if $A, B \in \mathcal{M}$, $A \subseteq B$, then $B \setminus A \in \mathcal{M}$, and if $(A_n) \subseteq \mathcal{M}$ with $A_n \subseteq A_{n+1}$ increasing then $\bigcup_{n \geq 1} A_n \in \mathcal{M}$.*

A collection of sets is a $\sigma$-algebra if and only if it is both a $\pi$-system and a $\lambda$-system.

**Lemma 1 ($\pi$-$\lambda$ systems lemma)** *Let $\mathcal{A}$ be a $\pi$-system and $\mathcal{M}$ a $\lambda$-system. Then if $\mathcal{A} \subseteq \mathcal{M}$ then $\sigma(\mathcal{A}) \subseteq \mathcal{M}$.*

We can use this with a convenient $\pi$-system to show that our $\lambda$-system contains more than is immediately obvious.

Let $\lambda(\mathcal{A})$ be the smallest $\lambda$-system containing $\mathcal{A}$. This is a subset of $\mathcal{M}$ and $\sigma(\mathcal{A})$, so we just need to show that $\lambda(\mathcal{A})$ is a $\sigma$-algebra (for which we just have to show that it is a $\pi$-system).

**Definition 5 (Random variables)** *With measurable spaces $(\Omega, \mathcal{F})$, $(E, \mathcal{E})$, a function $f : \Omega \to E$ is said to be an $E$-valued random variable (or a measurable function) if for all $A \in \mathcal{E}$, $f^{-1}(A) \in \mathcal{F}$.*

Note that this statement is equivalent to taking $f$ as a random variable in $(f(\Omega), \{A \cap f(\Omega) : A \in \mathcal{E}\})$.

We get immediately that random variables can be composed as one would expect. We can also use random variables to define new $\sigma$-algebras. Note that $(\Omega, \{f^{-1}(A) : A \in \mathcal{E}\})$ is a $\sigma$-algebra.

**Definition 6 (Generated $\sigma$-algebra)** *With $\{f_i : i \in I\}$ a family of functions $\Omega \to E$, $\sigma(f_i : i \in I)$ is the smallest $\sigma$-algebra on $\Omega$ for which all $f_i$ are measurable.*

This is initially a slightly intimidating definition, but the intuition is just that we need our $\sigma(f_i : i \in I) = \sigma(f_i^{-1}(A) : A \in \mathcal{E}, i \in I)$.

The following result is critical for our understanding of the measurability of random variables on these $\sigma$-algebras.

**Theorem 2 ($\sigma(X)$-measurability as functional dependence)** *With $X : (\Omega, \mathcal{F}) \to (E, \mathcal{E})$, $Y : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ random variables, $Y$ is $\sigma(X)$-measurable if and only if $Y = f \circ X$ for some random variable $f : (E, \mathcal{E}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$.*

To construct this we use a standard method of beginning with simple functions, and then using the density of $\mathcal{E}$-simple functions in $\mathcal{E}$-measurable functions under the pointwise topology, we can construct $f$ in general.

Noting that if $X_1, \ldots, X_n$ are measurable on $(\Omega, \mathcal{F})$, $(X_1, \ldots, X_n)$ is also measurable (start with showing the measurability property on the normal $\pi$-system generating $\mathcal{B}(\mathbb{R}^n)$, then use $\pi$-$\lambda$ systems lemma), this statement generalises much further than this, to include sequences of random variables as well.

**Theorem 3 (Monotone Class Theorem)** *Let $\mathcal{H}$ be a class of bounded functions from $\Omega \to \mathbb{R}$ such that*
- *$\mathcal{H}$ is a vector space over $\mathbb{R}$,*
- *the constant function $1 \in \mathcal{H}$,*
- *if $(f_n) \subseteq \mathcal{H}$, $f_n \to f$ monotonically increasing with $f$ bounded, then $f \in \mathcal{H}$,*

*then if $\mathcal{C} \subseteq \mathcal{H}$, and $\mathcal{C}$ is closed under pointwise multiplication, then all bounded $\sigma(\mathcal{C})$-measurable functions are in $\mathcal{H}$.*

To get an intuition for this, note that any $f \in \mathcal{C}$ is necessarily bounded and $\sigma(\mathcal{C})$-measurable, but the converse is not immediate. Thus we essentially get a statement of the $\lambda$-$\pi$ systems lemma but for functions on analogous systems.

We can firstly see that $\mathcal{H}$ is closed in $\mathcal{F}_b(\Omega)$. Then, we can prove the statement for the special case of $\mathcal{C} = \{\chi_A : A \in \mathcal{A}\}$ for a $\pi$-system $\mathcal{A}$, then adding 1 to $\mathcal{C}$ without loss of generality we can make the proof more general (concretely, because $\sigma(\mathcal{C}) \subseteq \sigma(\mathcal{C} \cup \{1\})$).

It may allow this theorem to make more sense to note that $\lambda$-systems are sometimes referred to as 'monotone classes'. Thus the $\pi$-$\lambda$ systems lemma can be seen as saying that for $\mathcal{A}$ a $\pi$-system, the (smallest) monotone class generated by $\mathcal{A}$ is $\sigma(\mathcal{A})$.

We can use the monotone class theorem to demonstrate that for $f : \Omega_1 \times \Omega_2 \to \mathbb{R}$ is measurable, then fixing $\omega_1 \in \Omega_1$, $\omega_2 \mapsto f(\omega_1, \omega_2)$ is measurable.

## Conditional Probability

Up until presently, we've considered the notion of event $A$ conditioned on event $B$ as having a fixed probability. This doesn't entirely capture what a conditional is however – we're conditioning on the amount of information we have, and therefore we want the conditional probability to change as a function of our information. In particular, we want our conditional probability to be a function of $\omega \in \Omega$, and in order to reflect conditioning as a reflection of information, we want to condition over events in a $\sigma$-algebra, rather than individual events.

Doing more algebra, we see that expectation is a more fitting operator, leading us to the following definition:

**Definition 7 (Conditional expectation)** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$, $\mathcal{G} \subseteq \mathcal{F}$ a $\sigma$-algebra. A random variable $Y \in \mathcal{L}^1(\Omega, \mathcal{G}, \mathbb{P})$ is (a version of) the conditional expectation of $X$ given $\mathcal{G}$ if for $G \in \mathcal{G}$,*

$$\mathbb{E}\big[Y\chi_G\big] = \mathbb{E}\big[X\chi_G\big].$$

Both aspects of this statement are key. $\mathcal{G}$-measurability entails that our conditional expectation is a random variable whose value provides no more information about $\mathcal{F}$ than is precisely contained in $\mathcal{G}$. That this random variable actually represents the expectation of $X$ subject to additional information is then represented by the equality of expectations on events in $\mathcal{G}$, often rewritten as:

$$\int_G \mathbb{E}\big[X \,|\, \mathcal{G}\big] \,\mathrm{d}\mathbb{P} = \int_G X \,\mathrm{d}\mathbb{P},$$

which allows us to carry over all of our normal integration properties to conditional expectations.

In particular, note that for random variables $Y_1, \ldots, Y_n$, $\mathbb{E}\big[X \,|\, \sigma(Y_1, \ldots, Y_n)\big] = F(Y_1, \ldots, Y_n)$ for some measurable $F : \mathbb{R}^n \to \mathbb{R}$, by representing measurability as functional dependence.

In order to retain our normal definition, one can see that for an event $A \in \mathcal{F}$, $\mathbb{E}\big[X \,|\, \sigma(A)\big]$ may be modified on a null set to be constant on the sets $\{A, A^c\}$. This allows us, in applied settings, to select arbitrarily for each $A \in \mathcal{F}$ some $\omega \in A$ so that

$$\mathbb{E}\big[X \,|\, A\big] = \mathbb{E}\big[X \,|\, \sigma(A)\big](\omega).$$

**Theorem 4** *The conditional expectation of $X$ given $\mathcal{G}$ exists, denoted $\mathbb{E}\big[X \,|\, \mathcal{G}\big]$, and if $Z$ is also the conditional expectation of $X$ given $\mathcal{G}$, then*

$$Z = \mathbb{E}\big[X \,|\, \mathcal{G}\big] \quad a.s.$$

While the uniqueness proof is relatively straightforward, the existence proof requires some methods from functional analysis. In particular, we prove the statement for $L^2(\Omega, \mathcal{F}, \mathbb{P})$, on noting that this space is Hilbert, and $L^2(\Omega, \mathcal{G}, \mathbb{P})$ is a closed subspace, allowing us to write any $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ as $X = Y + Z$ with $Y \in L^2(\Omega, \mathcal{G}, \mathbb{P})$, $Z \in L^2(\Omega, \mathcal{G}, \mathbb{P})$. We then extend this proof to $L^1$ by clamping the positive and negative parts of $X$ by $n$, and demonstrating by extending the MCT to conditionals that we can then define $Z$.

Note importantly how conditional expectations behave with respect to measurability. If we have $X$ a $\mathcal{G}$-measurable random variable, then

$$\mathbb{E}\big[X \,|\, \mathcal{G}\big] \overset{a.s.}{=} X.$$

Meanwhile if $\sigma(X)$ and $\mathcal{G}$ are independent, then

$$\mathbb{E}\big[X \,|\, \mathcal{G}\big] \overset{a.s.}{=} \mathbb{E}\big[X\big].$$

**Lemma 5 (Tower property)** *Take $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$, $\mathcal{F}_1$, $\mathcal{F}_2$ both $\sigma$-algebras, satisfying $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \mathcal{F}$. Then*

$$\mathbb{E}\Big[\mathbb{E}\big[X \,|\, \mathcal{F}_2\big] \,|\, \mathcal{F}_1\Big] \overset{a.s.}{=} \mathbb{E}\big[X \,|\, \mathcal{F}_1\big].$$

This should be relatively intuitive – $\mathbb{E}\big[X \,|\, \mathcal{F}_2\big]$ contains more information than can be represented in $\mathcal{F}_1$, but is fundamentally still expressing a reduced form of $X$, which can be reduced more to give $\mathbb{E}\big[X \,|\, \mathcal{F}_1\big]$.

One can also consider this as a commutativity statement: as $\mathbb{E}\big[X \,|\, \mathcal{F}_1\big]$ is $\mathcal{F}_2$-measurable, thus $\mathbb{E}\big[X \,|\, \mathcal{F}_1\big] = \mathbb{E}\big[\mathbb{E}[X \,|\, \mathcal{F}_1] \,|\, \mathcal{F}_2\big]$, so the tower property is stating that with $\mathcal{F}_1 \subseteq \mathcal{F}_2$:

$$\mathbb{E}\Big[\mathbb{E}\big[X \,|\, \mathcal{F}_1\big] \,|\, \mathcal{F}_2\Big] \overset{a.s.}{=} \mathbb{E}\Big[\mathbb{E}\big[X \,|\, \mathcal{F}_2\big] \,|\, \mathcal{F}_1\Big].$$

I'm tempted to claim the more general statement, that for $\mathcal{F}_1$, $\mathcal{F}_2$ both $\sigma$-algebras in $\mathcal{F}$:

$$\mathbb{E}\Big[\mathbb{E}\big[X \,|\, \mathcal{F}_1\big] \,|\, \mathcal{F}_2\Big] \overset{a.s.}{=} \mathbb{E}\big[X \,|\, \mathcal{F}_1 \cap \mathcal{F}_2\big].$$

This statement is true if $\mathcal{F}_1$ and $\mathcal{F}_2$ are independent, because then both sides are equal to $\mathbb{E}[X]$, but it seems that there could be a 'middle-ground' between independence and containment for which commutativity stops holding.

On the other side, attempting to prove this statement, the tripping point is that it's unclear that the LHS is $\mathcal{F}_1$-measurable (although clearly it is $\mathcal{F}_2$-measurable).

**Lemma 6** *Take $X$, $Y$ random variables on $(\Omega, \mathcal{F}, \mathbb{P})$ with $X$, $Y$, and $XY$ integrable. Then*

$$\mathbb{E}\big[XY \,|\, \sigma(Y)\big] \overset{a.s.}{=} Y\mathbb{E}\big[X \,|\, \sigma(Y)\big].$$

Ensure for yourself that it's clear why this implies the same holding for $\mathcal{G} \supseteq \sigma(Y)$ instead of $\sigma(Y)$.

## Measures on $\mathbb{R}$

**Definition 8 (Measure space)** *A measure space is a triple $(\Omega, \mathcal{F}, \mu)$ such that $\Omega$ is a set, $\mathcal{F}$ is a $\sigma$-algebra on $\Omega$, and $\mu : \mathcal{F} \to [0, \infty]$ is countably additive ($\mu$ is then a measure on $(\Omega, \mathcal{F})$).*

**Definition 9 (Distribution function)** *Let $\mu$ be a probability measure on $\mathcal{B}(\mathbb{R})$. The distribution function of $\mu$ is $F_\mu(x) = \mu(-\infty, x]$, where we require that $F_\mu$ is non-decreasing, tends to 0 as $x \to -\infty$, to 1 as $x \to \infty$, and is right continuous.*

**Definition 10 (Absolute continuity and equivalence)** *We say that $\nu$ is absolutely continuous with respect to $\mu$, $\nu \ll \mu$, if for any $A \in \mathcal{F}$, $\mu(A) = 0$ implies that $\nu(A) = 0$. Further, we say that $\mu$ and $\nu$ are equivalent, $\mu \sim \nu$, if $\mu \ll \nu$ and $\nu \ll \mu$.*

**Extensions**

For the most part, it's difficult to characterise a measure explicitly, due to $\sigma$-algebras being incredibly large in all but countable $\Omega$. We therefore wish to characterise them in terms of their value on algebras.

**Theorem 7 (Uniqueness of extension)** *Let $\mu_1$ and $\mu_2$ be measures on a space $(\Omega, \mathcal{F})$, and $\mathcal{A} \subseteq \mathcal{F}$ is a $\pi$-system with $\sigma(\mathcal{A}) = \mathcal{F}$. Then if $\mu_1(\Omega) = \mu_2(\Omega) < \infty$ and $\mu_1\big|_{\mathcal{A}} = \mu_2\big|_{\mathcal{A}}$, then $\mu_1 = \mu_2$.*

This follows immediately via the $\lambda$-$\pi$ systems lemma.

**Theorem 8 (Carathéodory Extension theorem)** *Let $\Omega$ be a set and $\mathcal{A}$ an algebra on $\Omega$, then with $\mu_0 : \mathcal{A} \to [0, \infty]$ a countably additive set function, there exists a measure $\mu : \sigma(\mathcal{A}) \to [0, \infty]$ such that $\mu\big|_{\mathcal{A}} = \mu_0$.*

One can derive this from defining the outer measure $\mu^*$ in terms of $\mu_0$, and claiming that a set is measurable iff for all $E \subseteq \Omega$, $\mu^*(E) = \mu^*(E \cap B) + \mu^*(E \setminus B)$. We can then prove that this gives the smallest $\sigma$-algebra containing $\mathcal{A}$.

Note additionally that the restriction of any measure to a smaller $\sigma$-algebra remains a measure.

**Lemma 9** *Let $(\Omega, \mathcal{F})$ be a measurable space and $(\mu_n)$ a sequence of probability measures on $\mathcal{F}$. With a sequence of positive numbers $(a_n)$ such that $\sum_{n=1}^{\infty} a_n = 1$,*

$$\mu := \sum_{n=1}^{\infty} a_n \mu_n$$

*is a probability measure on $(\Omega, \mathcal{F})$.*

This requires Fubini's theorem, which is given later on.

**Definition 11 (Distribution function)** *If a function $F : \mathbb{R} \to [0, 1]$ satisfies:*
*i. $F$ is non-decreasing;*
*ii. $F(x) \to 0$ as $x \to -\infty$, $F(x) \to 1$ as $x \to \infty$; and*
*iii. $F$ is continuous from the right,*
*then $F$ is a distribution function.*

**Theorem 10** *Let $F$ be a distribution function. Then there exists a unique Borel probability measure $\mu$ on $\mathbb{R}$ such that $\mu(-\infty, x] = F(x)$. Further, every Borel probability measure on $\mathbb{R}$ defines a distribution function.*

A corollary of this is that there is a unique Borel measure such that for all $a < b \in \mathbb{R}$, $\mu(a, b] = b - a$.

This result demonstrates that there is a bijection between measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and distribution functions. In particular, we call these measures the Lebesgue-Stieltjes measures.

The proof of this theorem follows using both of the extension theorems. In particular, we use the algebra of left open right closed intervals.

**Definition 12 (Pushforward measure)** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $X : \Omega \to E$, then for $A \in \mathcal{E}$,*

$$\mathbb{Q}(E) := \mathbb{P}(X^{-1}(E))$$

Thus we have a (very non-injective) map from random variables in $E$ to probability measures on $E$. This is useful, on the basis that many properties of random variables will just be properties of the pushforward measure rather than the random variable itself.

**Theorem 11** *For $\{(\Omega_i, \mathcal{F}_i, \mathbb{P}_i) : i \in \{1, \ldots, n\}\}$ a set of probability measures, there is a unique measure $\mathbb{P}$ on $\left(\prod_{i=1}^{n} \Omega_i, \bigtimes_{i=1}^{n} \mathcal{F}_i\right)$ such that for $E_i \in \mathcal{F}_i$ with $i \in \{1, \ldots, n\}$,*

$$\mathbb{P}\left(\prod_{i=1}^{n} E_i\right) = \prod_{i=1}^{n} \mathbb{P}_i(E_i).$$

It's hopefully natural here that one should aim an induction proof.

The theorem here then allows us to extend the matter to infinite products, although at this point we require that what we're dealing with are probability measures (to keep each term in $[0, 1]$ for convergence reasons), rather than just finite measures as could work with the previous statement of the theorem.

## Independence

**Definition 13 (Independence)** *With $(\Omega, \mathcal{F}, \mathbb{P})$ a probability space, $(\mathcal{G}_i)_{i=1}^n$ a collection of $\sigma$-algebras, these $\sigma$-algebras are independent if for $E_i \in \mathcal{G}_i$ for $i \in \{1, \ldots, n\}$*

$$\mathbb{P}\left(\bigcap_{i=1}^{n} E_i\right) = \prod_{i=1}^{n} \mathbb{P}(E_i).$$

*Further, an arbitrary collection $(G_i)_{i \in I}$ of $\sigma$-algebras is independent if any finite subset of the collection is independent.*

Note that this means $\{\varnothing, \Omega\}$ is independent of anything else.

Additionally, we say that a set $(X_i)_{i \in I}$ of random variables is independent iff $(\sigma(X_i))_{i \in I}$ is independent.

This definition requires a bit of work to deal with properly. One of the best general results we can attain quickly gives a fairly applicable result:

**Theorem 12** *With $(\Omega, \mathcal{F}, \mathbb{P})$ a probability space, $(\mathcal{A}_i)_{i \in I}$ an arbitrary collection of $\pi$-systems, then $(\sigma(\mathcal{A}_i))_{i \in I}$ are independent iff for any finite $J \subseteq I$, $A_i \in \mathcal{A}_i$ for $i \in J$:*

$$\mathbb{P}\left(\bigcap_{i \in J} A_i\right) = \prod_{i \in J} \mathbb{P}(A_i)$$

We also have the result that for any independent set of $\sigma$-algebras, any subset is also independent.

It takes a small bit of proving, but from the above results we get the lemma:

**Lemma 13** *With $(\Omega, \mathcal{F}, \mathbb{P})$, a family of independent random variables $X_i : \Omega \to E_i$, measurable functions $f_i : E_i \to \mathbb{R}$ for $i \in I$, then $(f(X_i))_{i \in I}$ are independent.*

**Tail events**

**Definition 14** *For a sequence of random variables $(X_n)$, the tail $\sigma$-algebra is defined as*

$$\mathcal{T} = \bigcap_{n=1}^{\infty} \sigma(\{X_k : k > n\})$$

The intuition here is that all events in the tail $\sigma$-algebra contain sample information distinguishing the results of functions of infinitely many sequence elements.

**Theorem 14 (Kolmogorov's 0-1 Law)** *Let $(X_n)$ be a sequence of independent random variables. Then the tail $\sigma$-algebra of $(X_n)$ contains only events with probability 0 or 1.*

To see this, we demonstrate that $\mathcal{T}$ is independent of a $\sigma$-algebra containing it, and therefore that all of its events are independent with themselves.

**Borel-Cantelli lemmas**

**Definition 15** *With $(A_n)$ a sequence of sets from $\mathcal{F}$:*

$$\begin{aligned}\limsup_{n \to \infty} A_n &= \bigcap_{n=1}^{\infty} \bigcup_{m \geq n} A_m \\ &= \{\omega \in \Omega : \omega \in A_n \text{ for infinitely many } n\} \\ &= \{A_n \text{ infinitely often}\}\end{aligned}$$

*and*

$$\begin{aligned}\liminf_{n \to \infty} A_n &= \bigcup_{n=1}^{\infty} \bigcap_{m \geq n} A_m \\ &= \{\omega \in \Omega : \omega \in A_n \text{ eventually }\} \\ &= \{A_n \text{ eventually}\}\end{aligned}$$

**Lemma 15 (Fatou and Reverse Fatou for sets)** *With $(A_n)$ a sequence of sets in $\mathcal{F}$,*

$$\begin{aligned}\mathbb{P}(\liminf_{n \to \infty} A_n) &\leq \liminf_{n \to \infty} \mathbb{P}(A_n) \\ \mathbb{P}(\limsup_{n \to \infty} A_n) &\geq \limsup_{n \to \infty} \mathbb{P}(A_n).\end{aligned}$$

**Lemma 16 (First Borel-Cantelli lemma)** *For $(A_n)$ a sequence of events in $\mathcal{F}$, if*

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty,$$

*then $\mathbb{P}(A_n \text{ i.o.}) = 0$.*

**Lemma 17 (Second Borel-Cantelli lemma)** *For $(A_n)$ a sequence of independent events in $\mathcal{F}$, if*

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty,$$

*then $\mathbb{P}(A_n \text{ i.o.}) = 1$.*

By their nature, the BC lemmas are only informative in relation to almost sure events. While this may seem incredibly limited, by Kolmogorov's 0-1 Law, it turns out that many events of interest are in fact almost sure events.

## Integration

As already covered in Part A Integration, we define integration as normal:

**Definition 16 (Integral on simple functions)** *For a measure space $(\Omega, \mathcal{F}, \mu)$, $f : \Omega \to [0, \infty]$ a non-negative simple function taking values $\{a_1, \ldots, a_n\} \subseteq \mathbb{R}$:*

$$\int f \,\mathrm{d}\mu = \sum_{i=1}^{n} a_i \mu\left(f^{-1}(\{a_i\})\right).$$

**Definition 17 (Integral on non-negative functions)** *For $f : \Omega \to [0, \infty]$ a non-negative measurable function:*

$$\int f = \sup\left\{\int g \,\mathrm{d}\mu : g \text{ simple}, 0 \leq g \leq f\right\}$$

**Definition 18 (Integral)** *For $f : \Omega \to \mathbb{R}$ a measurable function and $\int |f| \,\mathrm{d}\mu < \infty$, we write $f^+ = \max(f, 0)$, $f^- = -\min(f, 0)$, and*

$$\int f \,\mathrm{d}\mu = \int f^+ \,\mathrm{d}\mu - \int f^- \,\mathrm{d}\mu$$

**Theorem 18 (Monotone convergence theorem)** *For $(f_n)$ a sequence of non-negative functions measurable on $(\Omega, \mathcal{F}, \mu)$, such that $f_n \to f$ monotonically. Then*

$$\int f_n \,\mathrm{d}\mu \to \int f \,\mathrm{d}\mu.$$

**Theorem 19 (Fatou's lemma)** *For $(f_n)$ a sequence of non-negative functions measurable on $(\Omega, \mathcal{F}, \mu)$,*

$$\int \liminf_{n \to \infty} f_n \,\mathrm{d}\mu \leq \liminf_{n \to \infty} \int f_n \,\mathrm{d}\mu$$

**Lemma 20 (Reverse Fatou's lemma)** *For $(f_n)$ a sequence of non-negative functions measurable on $(\Omega, \mathcal{F}, \mu)$, assume that there is an integrable function $g$ such that $f_n \leq g$ for $n \geq 1$. Then*

$$\int \limsup_{n \to \infty} f_n \,\mathrm{d}\mu \geq \limsup_{n \to \infty} \int f_n \,\mathrm{d}\mu$$

**Theorem 21 (Dominated convergence theorem)** *For $(f_n)$ a sequence of functions measurable on $(\Omega, \mathcal{F}, \mu)$ with $f_n \to f$ pointwise. Assume that there is an integrable function $g$ such that $|f_n| \leq g$ for $n \geq 1$. Then*

$$\int f_n \,\mathrm{d}\mu \to \int f \,\mathrm{d}\mu.$$

**Theorem 22** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $X : \Omega \to E$, $g : E \to \mathbb{R}$ measurable on their respective spaces. Then $g$ is $(\mathbb{P} \circ X^{-1})$-integrable iff $g \circ X$ is $\mathbb{P}$-integrable. Further,*

$$\int_E g(x) \,\mathrm{d}(\mathbb{P} \circ X^{-1})(x) = \int_\Omega g(X(\omega)) \,\mathrm{d}\mathbb{P}(\omega).$$

**Definition 19 (Expectation)** *For $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$,*

$$\mathbb{E}[X] = \int_\Omega X \,\mathrm{d}\mathbb{P}.$$

We say that $X$ admits an $n$th moment if $X \in \mathcal{L}^n(\Omega, \mathcal{F}, \mathbb{P})$.

**Definition 20 (Variance)** *For $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$,*

$$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right].$$

**Definition 21 (Standardised moment)** *If $X \in \mathcal{L}^n(\Omega, \mathcal{F}, \mathbb{P})$, the $n$th standardised moment of $X$ is*

$$\mathbb{E}\left[\left(\frac{X - \mathbb{E}[X]}{\sqrt{\mathrm{Var}(X)}}\right)^n\right]$$

**Theorem 23 (Fubini-Tonelli)** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be the product of probability spaces $(\Omega_i, \mathcal{F}_i, \mathbb{P}_i)$ for $i \in \{1, 2\}$, and $f : \Omega \to \mathbb{R}$ is a bounded measurable function. Then both*

$$x \mapsto \int_{\Omega_2} f(x, y) \,\mathrm{d}\mathbb{P}_2(y)$$

$$\text{and} \quad y \mapsto \int_{\Omega_1} f(x, y) \,\mathrm{d}\mathbb{P}_1(x)$$

*are measurable (respectively in $\mathcal{F}_1$ and $\mathcal{F}_2$).*

*If either $f \geq 0$ or $f$ is $\mathbb{P}$-integrable over $\Omega$, then*

$$\int_\Omega f \,\mathrm{d}\mathbb{P} = \int_{\Omega_1} \int_{\Omega_2} f(x, y) \,\mathrm{d}\mathbb{P}_1(x) \,\mathrm{d}\mathbb{P}_2(y) = \int_{\Omega_2} \int_{\Omega_1} f(x, y) \,\mathrm{d}\mathbb{P}_2(y) \,\mathrm{d}\mathbb{P}_1(x)$$

**Radon-Nikodym theorem**

Integration as we've defined it gives a canonical method of defining a measure on a space: as the integral of a fixed non-negative measurable function over the set being measured.

Concretely: with a measure space $(\Omega, \mathcal{F}, \mu)$, a measurable function $f : \Omega \to [0, \infty]$, $A \in \mathcal{F}$,

$$\nu(A) := \int_A f \,\mathrm{d}\mu$$

is a measure on $\mathcal{F}$ (via MCT).

We therefore want to characterise how often a measure can be characterised in this way (in particular, whether we can get this the case with respect to leb or the counting measure, both for which we have a wealth of tools).

**Theorem 24 (Radon-Nikodym theorem)** *Let $\mu$, $\nu$ be two probability measures on a $\sigma$-algebra $(\Omega, \mathcal{F})$. Then $\nu \ll \mu$ if and only if there is a measurable function $f : \Omega \to [0, \infty]$ such that for $A \in \mathcal{F}$,*

$$\nu(A) = \int_A f \,\mathrm{d}\mu.$$

*Further, $\nu \sim \mu$ if and only if $\mu(f^{-1}(\{0\})) = \nu(f^{-1}(\{0\})) = 0$.*

We call $f$ the radon-nikodym derivative of $\nu$ with respect to $\mu$, $f = \frac{\mathrm{d}\nu}{\mathrm{d}\mu}$. If $\nu \sim \mu$, then $\frac{1}{f} = \frac{\mathrm{d}\mu}{\mathrm{d}\nu}$.

This means that providing $\mathrm{leb}(A) = 0$ implies that $\nu(A) = 0$, we can construct $\nu$ in this way.

Using this theorem, we can define for $A, B \in \mathcal{F}$ the conditional distribution $\mathbb{P}(A \,|\, B)$, provided $\mathbb{P}(B) > 0$. If $\mathbb{P}(A) = 0$, then $\frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = 0$, so there is some $f_B : \Omega \to [0, \infty]$ measurable such that

$$\frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \int_A f_B(\omega) \,\mathrm{d}\mathbb{P}(\omega).$$

## Uniform Integrability

We've now introduced the notions of convergence in $\mathcal{L}^1$, and seen that it implies convergence in probability. We'd quite like to go the other direction however, and find a sufficient condition for which the two are the same.

**Definition 22 (Uniform integrability)** *A collection $\mathcal{C}$ of random variables is called uniformly integrable (UI) if*

$$\lim_{N \to \infty} \sup_{X \in \mathcal{C}} \mathbb{E}[|X|\chi_{|X| > N}] = 0$$

Importantly, this is a property of collections, rather than individual random variables. The larger our collection is, the less likely it is to hold (and conversely, if $\mathcal{C}$ is UI, then $\mathcal{D} \subseteq \mathcal{C}$ is also UI). We can see immediately therefore that this is a property that can only hold for collections of random variables in $\mathcal{L}^1$, by considering the singleton sets which can be UI.

If we have a $Y \in \mathcal{L}^1$ such that for $X \in \mathcal{C}$, $|X| \leq Y$, then $\mathcal{C}$ is uniformly integrable. This is one of the most common methods for demonstrating a collection is UI.

Another useful characterisation is found by reframing $|X|\chi_{|X| \geq N}$. From the below fact, we can see that we can replace it with $(|X| - N)^+$ in the definition of uniform integrability without effect:

$$0 \leq (|X| - N)^+ \leq |X|\chi_{|X| \geq N} \leq 2(|X| - N/2)^+.$$

A common formulation that allows us to avoid having to determine the sets on which $|X| \geq N$ is given below:

**Lemma 25** *Let $\mathcal{C}$ be a family of random variables. Then $\mathcal{C}$ is UI iff $\sup_{X \in \mathcal{C}} \mathbb{E}[|X|] < \infty$ and*

$$\sup\left\{\mathbb{E}\big[|X|\chi_A\big] : X \in \mathcal{C},\ A \in \mathcal{F},\ \mathbb{P}(A) \leq \delta\right\} \to 0$$

*as $\delta \to 0$.*

**Theorem 26 (Vitali's convergence theorem)** *Take $(X_n)$ a sequence of integrable random variables which converge in probability to a random variable $X$. The following are equivalent:*
*i. $\{X_n : n \geq 1\}$ is uniformly integrable.*
*ii. $X \in \mathcal{L}^1$ and $\mathbb{E}[|X_n - X|] \to 0$ as $n \to \infty$.*
*iii. $X \in \mathcal{L}^1$ and $\mathbb{E}[|X_n|] \to \mathbb{E}[|X|]$ as $n \to \infty$.*

It's worth noting that (ii) is the definition of convergence in $\mathcal{L}^1$, from which we get (iii) just via the reverse triangle inequality. The difficult part of this proof is demonstrating that (iii) entails (i) which requires that characterisation of UI in terms of $(|X| - N)^+$ in conjunction with convergence in probability.

By a non-examinable result (Dunford-Pettis), a collection is UI iff its closure is compact in $\sigma(L^1, L^\infty)$, which is the weak topology on $L^1$.

**Theorem 27** *Take $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$, $\{\mathcal{F}_i : i \in I\}$ a family of $\sigma$-algebras in $\mathcal{F}$. Then $\big\{\mathbb{E}\big[X \,|\, \mathcal{F}_i\big] : i \in I\big\}$ is uniformly integrable.*

This result follows via Jensen's inequality, and provides a critical example of UI families – particularly useful for considering backwards martingales later on.

## Filtrations

**Definition 23 (Filtrations)** *For $(\Omega, \mathcal{F}, \mathbb{P})$ a probability space, a filtration $(\mathcal{F}_n)$ is an sequence of $\sigma$-algebras ($\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$ for $n \geq 1$). We then call $(\Omega, \mathcal{F}, (\mathcal{F}_n), \mathbb{P})$ a filtered probability space.*

**Definition 24 (Adapted stochastic process)** *Take $(X_n)$ a sequence of random variables, $(\mathcal{F}_n)$ a filtration. If for $n \geq 1$, $X_n$ is $\mathcal{F}_n$ measurable, then $(X_n)$ is said to be adapted to $(\mathcal{F}_n)$.*

Note that for a stochastic process $(X_n)$, we get a natural filtration

$$\mathcal{F}_n := \sigma\left(\{X_k : k \leq n\}\right).$$

This is the smallest filtration possible.

It's worth noting that with $\mathcal{F}_\infty = \bigcup_{n \geq 1} \mathcal{F}_n$, we don't necessarily have that $\mathcal{F}_\infty = \mathcal{F}$. Nonetheless, once we're working primarily within the filtered space, this shouldn't make a significant amount of difference, because the remainder of events will be inaccessible to any adapted process.

**Definition 25 (Stopping time)** *Take $(\Omega, \mathcal{F}, (\mathcal{F}_n), \mathbb{P})$ a filtered probability space. A random variable $\tau : \Omega \to \mathbb{N} \cup \{\infty\}$ is a stopping time with respect to $(\mathcal{F}_n)$ if for $n \geq 1$, $\tau^{-1}(\{n\}) \in \mathcal{F}_n$.*

Immediate examples of stopping times can be given. Any constant is a stopping time, as are the maxima or minima of any two stopping times. Further, the first hitting time for an adapted stochastic process on any measurable set is a stopping time.

**Definition 26** *Let $\tau$ be a stopping time on $(\Omega, \mathcal{F}, (\mathcal{F}_n), \mathbb{P})$. The $\sigma$-algebra of information at time $\tau$ is*

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